Lecture 5

Tuesday, 24 January 2023 10:30 AM

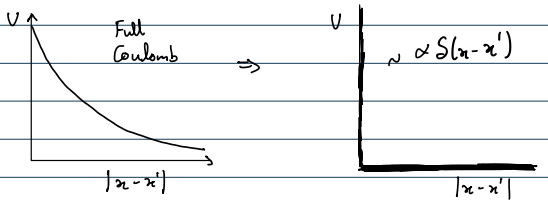
- Start thinking of term projects → List 2-3 thoughts
- Tutorial session → next thursday

One-dimensional Atom

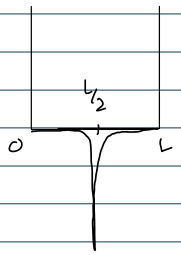
$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int \rho^3\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\iint \frac{\rho(x)\rho(x')}{|x-x'|}\,dx\,dx' - \underbrace{C_x\frac{e^2}{4\pi\varepsilon_0}\int \rho^2\,dx}_{\text{Exchange energy}} + \int V_{ext}(x)\,\rho\,dx$$

Simplify Coulomb terms using delta functions:

$e^- e^-$ repulsion



$$V_{ext} = -\gamma\,\delta\left(x-\frac{L}{2}\right)$$



From an e^- perspective, nucleus is external.

Thus, our 1D atom becomes,

($e^- e^-$ interactions; e^- nucleus interaction)

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_0^L \rho^3\,dx - C_x\frac{e^2}{4\pi\varepsilon_0}\int \rho^2\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int_0^L\int_0^L \delta(x-x')\rho(x)\rho'(x)\,dx'\,dx - \gamma\int_0^L \delta\left(x-\frac{L}{2}\right)\rho\,dx$$

$$= \frac{\pi^2\hbar^2}{6m}\int_0^L \rho^3\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int_0^L [\rho(x)]^2\,dx - C_x\frac{e^2}{4\pi\varepsilon_0}\int_0^L \rho^2\,dx - \gamma\,\rho\left(\frac{L}{2}\right) - \mu\left(\int_0^L \rho\,dx - N\right)$$

$$\frac{\delta E}{\delta \rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + 2\frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{2} - C_x\right)\rho_{min} - \mu$$

($\frac{1}{2}$: Coulomb; C_x: exchange)

ρ_{min} = constant →

$$\int_0^L \rho_{min}\,dx = N$$

$$\rho_{min} = \frac{N}{L}$$

↙ Plugging back into energy functional

Electron Density

$$E_{\ldots}[\rho] = \frac{\pi^2\hbar^2}{\ldots}N^3 + \frac{e^2}{\ldots}\left(1 - C_x\right)N^2 - \frac{\gamma}{\ldots}N$$

Electron Density

energy functional

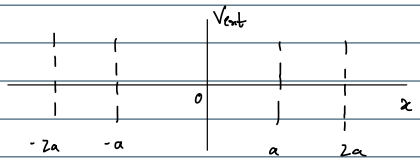
$$E_{GS} = E[\rho] = \underbrace{\frac{\pi^2\hbar^2}{6mL^2}N^3}_{KE} + \underbrace{\frac{e^2}{4\pi\varepsilon_0 L}\left(\frac{1}{2} - C_m\right)N^2}_{\text{interaction term}} - \underbrace{\frac{\gamma}{L}N}_{\text{attractions to nucleus}}$$

Note: These treatments work okay for atoms but not for molecules

DFT & TISE w/ periodic boundary conditions

eg: hot catalysis might have some periodicity

1) Basic idea: $V_{ext}(x) = V_{ext}(x+a)$



If $V_{ext}(x)$ is periodic, observable properties are also periodic: $\rho(x) = \rho(x+a)$

Approach $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_{-na}^{+na}\rho^3\,dx + \int_{-na}^{na}V_{ext}(x)\,\rho\,dx$; n = some integer

↳ $\frac{\delta F}{\delta\rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + V_{ext}(x) - \mu$

$\rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}\left(\mu - V_{ext}(x)\right)\right)^{1/2}$ → periodic in x ✓

Cx: Dirac Comb:



$V_{ext}(x) = \alpha\sum\delta(x-na)$; n is a positive or negative integer

P periods are analyzed

$$E[\rho] = \int_{-\frac{Pa}{2}}^{\frac{Pa}{2}}\frac{\pi^2\hbar^2}{6m}\rho^3\,dx + \int_{-\frac{Pa}{2}}^{\frac{Pa}{2}}\alpha\sum\delta(x-na)\,\rho\,dx$$

$\frac{\delta F}{\delta\rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 - \mu$; $\rho_{min} = \text{constant} = \frac{N}{L}$ (N is total # of particles in all P periods)

$E[\rho_{min}] = \frac{\pi^2\hbar^2}{6m}\frac{N^3}{(Pa)^2} + \alpha\,\rho(na)\,P$; Let $N_{cell} = \frac{N}{P}$

($\rho(na) \to \frac{N}{L}$)

$$\boxed{\frac{E_{GS}}{P} = \frac{\pi^2\hbar^2}{6ma^2}N_{cell}^3 + \frac{\alpha}{a}N_{cell}}$$

energy per unit cell in a Dirac potential

Kohn-Sham DFT

+ shares many features w/ TISE treatment of non-interacting particles ⇒ Need to study periodic treatment of TISE

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+ As w/ DFT, assume $V(x) = V(x+na)$ n = integer

+ All measurable quantities are also periodic $\rho(x) = \rho(x+a)$

$$|\Psi(x)|^2 = \Psi^*\Psi = |\Psi(x+a)|^2 \quad ; \text{ wavefunctions are complex so } \Psi(x) \neq \Psi(x+a)$$

+ Most general periodicity is: $\Psi(x+a) = \lambda\Psi(x)$ w/ $\lambda^*\lambda = 1$

+ If we have P total periods, and we assume periodic BC's (the system "wraps around" itself)

$$\Psi(x+Pa) = \lambda^P\Psi(x) = \Psi(x) \quad ; \quad \lambda^P = 1 = e^{2n\pi i} = \cos(2n\pi) + i\sin(2n\pi) \quad ; \quad n = 0, \pm1, \pm2, \ldots$$

$$\lambda = e^{\frac{2\pi n}{P}i} = e^{ika} \quad ; \quad k = \frac{2\pi n}{Pa}$$

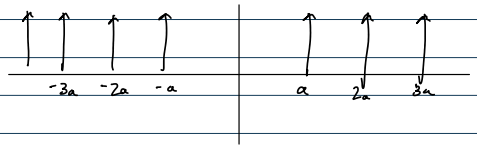
(k: wave number; Pa: total length of system)

Wavefunction displaced by a period →

$$\therefore \underset{k}{\Psi}(x+a) = \Psi_k(x)e^{ika} \quad ; \quad k = \frac{2\pi n}{Pa} \quad ; \quad n = 0, \pm1, \pm2, \ldots \quad ; \quad k \text{ is quantum number depending on periodicity}$$

} Bloch's Theorem

Ex: Dirac Comb:



For $0 < x < a$, $V(x) = 0$; so TISE is $-\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = E\Psi$

$$\frac{d^2\Psi}{dx^2} = -k^2\Psi \quad ; \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\Psi(x) = A\sin(kx) + B\cos(kx) \quad ; \quad 0 < x < a$$

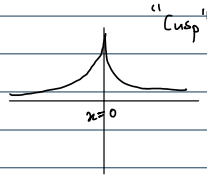
From Bloch's theorem for $-a < x < 0$,

$$\Psi(x+a) = e^{ika}\left[A\sin(k(x+a)) + B\cos(k(x+a))\right]$$

Initial conditions: @ $x = 0$, $\Psi_{0^-} = \Psi_{0^+}$ (Ψ is cont.)

$$\hookrightarrow B = e^{-ika}\left[A\sin(ka) + B\cos(ka)\right]$$

Also, $\left.\frac{d\Psi}{dx}\right|_{0^+} - \left.\frac{d\Psi}{dx}\right|_{0^-} = \frac{2m\alpha}{\hbar^2}\Psi(0)$



$$kA - e^{ika}k\left[A\cos(ka) - B\sin(ka)\right] = \frac{2m\alpha}{\hbar^2}B$$

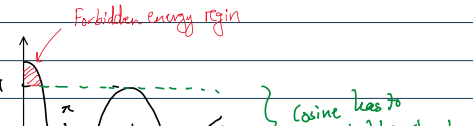
Eliminate A & B:

$$\boxed{\cos(Ka) = \cos(ka) + \frac{m\alpha a}{\hbar^2}\frac{\sin(ka)}{ka}}$$

Solve for k given K:

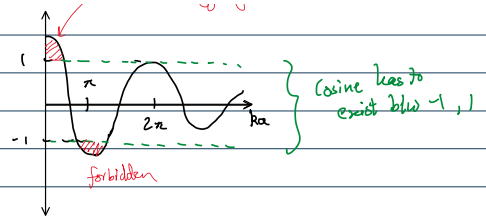
(graphically)

For $\frac{m\alpha a}{\hbar^2} = 1$



For $\frac{mVa}{\hbar^2} = 1$

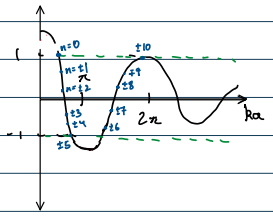
$K = \frac{2\pi n}{Pa}$



Certain values of $k (\sim E^{1/2})$ are forbidden by periodicity.
This causes band structure in periodic systems

K is quantized, as $P \to \infty$ K becomes continuous

↳ Try P = 10



States for P+1 non-interacting fermions in first band (ignoring spin) and P states for all other bands

To get total E, sum energies of particles (non-interacting)

$$E_{tot} = \sum_{\text{particles}} \varepsilon_n = \sum_{\text{parti}} \frac{\hbar^2}{2m} k_n^2 \quad ; \text{ If N is large we can approximate sum w/ integral}$$

$$= \sum \frac{\hbar^2}{2m} k_n^2 \Delta n \quad ; \quad \Delta n = \frac{Pa}{2\pi} \Delta K$$

$$= \frac{Pa}{2\pi} \sum \frac{\hbar^2}{2m} k_n^2 \Delta K \approx \frac{Pa}{2\pi} \int \frac{\hbar^2}{2m} [k(K)]^2 \cdot dK$$

$$\boxed{\frac{E_{tot}}{P} \approx \frac{a}{2\pi} \int_{\substack{\text{occupied} \\ \text{bands}}} \frac{\hbar^2}{2m} [k(K)]^2 \, dK}$$

Energy per cell in periodic system of non-interacting particles ($P \to \infty$) K is cont.